Model	P	R	F1	Acc
Finetuned on (ANLI +ALL)				
DistilBERT (Sanh et al., 2019)	59.55	67.30	63.13	77.34
XLNet base(Yang et al., 2019)	63.16	75.47	68.75	75.72
RoBERTa base (Liu et al., 2019)	63.15	75.63	68.90	78.53
XLNet Large (Yang et al., 2019)	63.26	76.11	69.22	79.20
RoBERTa large (Liu et al., 2019)	**65.32**	**78.03**	**71.14**	**81.34**
Trained on our dataset				
DistilBERT	69.50	70.69	70.07	87.70
XLNet base	71.13	75.68	73.05	88.07
RoBERTa base	74.49	72.43	73.40	89.83
XLNet large	74.41	**80.00**	76.76	89.64
RoBERTa large	**79.17**	76.17	**77.55**	**91.50**

Table 2: Results of Contradiction Classifier; P: Precision, R: Recall, F1: F1-Score, Acc: Accuracy

be attributed to the differences in reviews and the content of the existing datasets. Scientific peer review comments are written differently from typical human-written English sentences and Wikipedia entries. They contain a more technical style of writing, which is challenging for models to parse when trained on current datasets. Our findings, therefore, highlight the pressing need for innovative datasets in this specific domain. We discuss the results of our proposed intermediate aspect sentiment model in Section D.

Next, we evaluate the performance of RoBERTa Large across the entire process (in evaluation mode). We obtain an accuracy of 88.60% from the Aspect Sentiment Classifier (in determining whether a pair of reviews has any SDAP or not) and a 74.25 F1 score for the Reviewer Disagreement Classifier. We compare our findings with those achieved by the zero-shot Large Language Model, ChatGPT. On the test set, ChatGPT scored an F1 of 64.67 which is 9 points lower than the baseline model, likely due to its lack of explicit training for this specific downstream task. We discuss the prompts and outputs in details in the Appendix E.

We also discuss where our proposed baseline fails (Error Analysis) in the Appendix F. We also utilized the BARD API [2] for our evaluation. We provided identical prompts to both Google BARD and CHATGPT to ensure a fair comparison. We compared our findings with those achieved by Google BARD. BARD scored an F1 of 61.35 on the test set, 12 points lower than the baseline model. This is likely due to its BARD's requirement for more specialized training for this specific task. We found that Google BARD registered a higher number of false positives compared to CHATGPT.

[2] https://github.com/dsdanielpark/Bard-API

7 Conclusion and Future Work

In this work, we introduce a novel task to automatically identify contradictions in reviewers' comments. We develop ContraSciView, a review-pair contradiction dataset. We designed a baseline model that combines the MIMLLN framework and NLI model to detect contradictory review pair comments. We found that RoBERTa large performed best for this task with F1 score of 71.14.

Our proposed framework doesn't consider the full review context when predicting contradictions. In the future, we will investigate a system that takes into account the entire review context while detecting contradictory review pair comments. Additionally, we plan to expand our dataset to include other domains and explore the significance of Natural Language Inference for the given task. We also aim to categorize contradictions based on their severity: high, medium, or low.

Limitations

Our study mainly focuses on identifying "explicit" contradictions. **Explicit Contradictions:** These are clear, direct, and unmistakable contradictions that can be easily identified. For example: The author claims in the introduction that "X has been proven to be beneficial," but in the discussion section, they state that "X has not been shown to have any benefit." One reviewer says, "Figure 2 clearly shows the relationship between A and B," while another reviewer comments, "Figure 2 does not show any clear relationship between A and B."

We do not delve into "implicit" contradictions, which can be hard to detect and can be subjective, making them a topic of debate. **Implicit Contradictions:** These are more subtle and may require deeper understanding or closer examination to identify. It may require the annotators to read the paper and also learn many related works or things to annotate. They are not directly stated but can be inferred from the context or how information is presented. For example: Review 1: "the method lacks algorithmic novelty and the exposition of the method severely inhibits the reader from understanding the proposed idea ." Review 2: "the work presented is novel but there are some notable omissions - there are no specific numbers presented to back up the improvement claims graphs are presented but not specific numeric results - there is limited discussion of the computational cost of the framework presented - there is no comparison to a baseline in which the additional learning cycles used for learning the embedding are used for training the student model ." For instance, Review1 mentions that the method lacks algorithmic novelty, and Review2 acknowledges the work as a novel but points out some notable omissions. This difference in perception is not a direct contradiction, as one reviewer finds the method lacking in novelty, while the other recognizes it as novel but with some omissions.

Additionally, we do not incorporate information from the papers being reviewed, as they cover a wide range of topics and would require many experts from various domains. However, we acknowledge that finding a method to uncover these implicit contradictions in reviews is an intriguing opportunity for future research.

Ethics Statement

We have utilized the open source peer review dataset for our work. The system sometimes generate incorrect contradictions or overlook certain contradictions. Like other general AI models, this model is not 100% accurate. Editors or Chairs might rely on more than just the contradictions predicted by the model to make decisions. It is important to emphasise that the primary purpose of this system is to assist editors by highlighting potential contradictions between reviewers; this system is only for editors' internal usage, not for authors or reviewers. Especially given their often busy schedules and the myriad of decisions they must make. The contradictions among the reviewers will help editors to identify and initiate a discussion to resolve the confusion between the reviewers and to make an informed decision. Since reviews are frequently extensive and intricate, it is challenging for editors to scrutinise every comment and address conflicts. This system aims to aid them in spotting such contradictions. For instance, if one comment reads, "The paper does not have any new findings," and another reviewer mentions, "The paper is somewhat novel," the editor might not immediately perceive this as contradictory. However, understanding the context and intent behind these comments is vital. If they represent a contradiction, editors can address it using the standard guidelines. The system does not detect all contradictions. If the system fails to identify a contradiction, it does not automatically mean none exists. Given its nature as a general AI system, there is a possibility of it presenting false negatives. Editors relying solely on this system could impact the review process. Any contradictions spotted outside the system's recommendations should also be addressed according to the guidelines. As an AI based model this is prone to errors, the editor/chair are advised to utilize this tool only for assistance and verify the contradiction and analyze carefully before making a decision.

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